

JOSEPH EZEKIEL

Mechatronics Engineer, Controls & Automation

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📍 Lagos, Nigeria 🏠 Open to relocation to Calgary, AB

Mechatronics engineer with hands-on industrial automation experience and a strong software/systems background. Experienced with PLC programming, SCADA/HMI, VFDs, instrumentation, electrical controls, P&IDs, Linux, networking, databases, and production software systems. Interested in control systems, industrial communications, reliability, and automation engineering.

EDUCATION

Federal University of Agriculture, Abeokuta, B.Eng. Mechatronics Engineering · 4.19/5.00 Mar 2017 – Mar 2023 | Ogun, Nigeria
Final-Year Project: Designed and built a model of a **PLC-based automated car wash system**, integrating sequencing, sensors, actuation, and control logic.

CONTROLS, ELECTRICAL & INDUSTRIAL EXPERIENCE

Industrial Automation Trainee, Kimberly Engineering Limited May 2021 – Oct 2021 | Lagos, Nigeria

- Programmed, interfaced, and troubleshoot **Siemens S7-300/400** and **Allen-Bradley PLCs** using STEP 7 and RSLogix 500 in hands-on lab and factory environments.
- Designed and configured **SCADA/HMI** interfaces with WinCC Flexible and FactoryTalk ME, including tags, alarms, controls, and operator displays.
- Configured and commissioned **Siemens Micromaster** and **Allen-Bradley PowerFlex VFDs**; worked with instrumentation, sensors, relays, contactors, motor starters, and process-control loops.
- Read and worked from **P&IDs and electrical/control schematics**, supported control-panel wiring and troubleshooting, and participated in factory automation work.
- Assisted with procurement of automation and electrical components and delivered selected practical training sessions to junior trainees.

Independent Solar Power Technician, Self-employed Jul 2024 – Sep 2024 | Lagos, Nigeria

- Assessed customer sites and designed solar backup systems using **load calculations, battery/inverter sizing, and component selection** based on expected demand and runtime requirements.
- Installed and commissioned solar panels, batteries, inverters, and charge controllers; completed system wiring, configuration, testing, and troubleshooting for residential backup-power systems.

Technical Intern, MDS Logistics Ltd Feb 2020 – Mar 2020 | Lagos, Nigeria

- Supported tractor-trailer maintenance and repair, including diesel engines and air-compression braking systems.
- Performed diesel quality testing and gained practical exposure to **arc and gas welding** and mechanical maintenance work.

SOFTWARE SYSTEMS EXPERINCE

Backend & Platform Engineer, Retreev Jun 2026 – Present

- Building backend, image-indexing, and production infrastructure for an event-photo platform; owning architecture, deployment, observability, and service reliability.
- Develop distributed services in **Go and Python**, working with PostgreSQL, Redis, Linux, Docker, networking, and internal service communication.

Software Engineer, Verboheit Oct 2024 – May 2026

- Built and operated backend systems for a national mathematics competition, including exams, scoring, staff workflows, notifications, and real-time services.
- Provisioned Linux environments, automated deployments and backups, and implemented monitoring and distributed tracing for production systems.

TEACHING EXPERIENCE

Physics Teacher, National Youth Service Corps Jul 2023 – Jun 2024 | Rivers, Nigeria

- Taught senior secondary students foundational physics, translating mathematical and engineering concepts into clear practical explanations.
- Managed classroom instruction, assessments, and student reporting while working within a structured school environment.

CERTIFICATIONS

Certified Industrial Automation Engineer • IBM DevOps and Software Engineering Professional • IBM Back-End Development Professional

TECHNICAL SKILLS

Controls & Automation: PLC programming, Siemens S7-300/400, Allen-Bradley RSLogix 500, SCADA/HMI, WinCC Flexible, FactoryTalk ME, VFDs, instrumentation, process control, electrical troubleshooting, P&IDs • **Engineering Tools:** Proteus, PTC Creo • **Software & Systems:** Go, Python, TypeScript, SQL, PostgreSQL, Redis, Linux, Docker, Git, networking, observability

SELECTED PROJECTS

Appa, Self-hosted Deployment Platform

- Built a mini-PaaS whose CLI provisions and hardens VPS fleets using Ansible, then deploys applications from Git, ZIP, or local source using Railpack, BuildKit, and Caddy with automatic HTTPS.
- Engineered idempotent infrastructure playbooks and a WebSocket live-log system with a single goroutine coordinating broadcast state through channels to avoid lock contention during concurrent deployments.

Resumable Upload Server

Implemented a resumable file-upload server in Go based on the tus protocol, handling upload creation, offset validation, incremental PATCH requests, recovery from interrupted transfers, and protocol-compliant HTTP behavior.

Paystack Python SDK

Built and published a Python wrapper for Paystack APIs using TDD, achieving ~99% test coverage with automated CI and Codecov reporting.

PROFESSIONAL AFFILIATION

Nigerian Society of Engineers, Graduate Member